Desk Research

BirdWatch NL — AI-Powered Bird Identification and Monitoring Assistant

Country	The Netherlands
Hackathon	AI for Good — Hackathon 7: Eyes on Earth
Project idea	A simple photo-based tool that detects and counts selected common bird groups, lets users correct the AI, and saves their observations.
Required tool	Roboflow for the custom computer-vision model; Python and Streamlit for the app.
Primary SDG	SDG 15 — Life on Land
Primary users	Beginner birdwatchers, students and local nature enthusiasts.

The central idea

BirdWatch NL starts with a familiar experience: you see birds beside a canal or in a park, take a photo, and then wonder what you are looking at and how many birds are in the picture. The project tests whether a small AI tool can make that first step easier. It is an educational prototype, not a replacement for trained bird counters or a system that can prove whether bird populations are recovering.

1. Problem statement

Background: why bird observations matter

Birds are often one of the easiest parts of nature to notice in everyday life. We see them on canals, near ponds, in gardens and in city parks. But seeing a bird and recording a useful observation are not quite the same thing. A beginner might recognise a pigeon or a mallard, but feel less certain when birds are far away, partly hidden, seen from an unusual angle, or mixed together in one photograph. It can also be surprisingly easy to lose track of a sighting if the date, place or number of birds is not written down.

There is a wider reason to pay attention. Bird populations in the Netherlands do not all follow the same trend, and it would be misleading to say that every bird species is declining. The Compendium voor de Leefomgeving reports that the average population trend for 184 breeding-bird species increased by 26% between 1990 and 2025. However, that overall figure hides important differences: 82 species declined, while 88 increased. In agricultural areas, the situation is more worrying; in a group of 27 characteristic farmland-bird species, 21 declined between 1990 and 2024 (Compendium voor de Leefomgeving, 2026a, 2026b). This is a reminder that we need reliable observations to understand what is happening to different bird groups and habitats.

The Netherlands already has well-established ways to collect those observations. Sovon Vogelonderzoek Nederland coordinates bird surveys that are used to understand population trends and support nature policy. Its waterbird monitoring programme includes monthly counts from September to April and contributes to national and international population estimates. Almost 2,000 volunteers and professionals contribute to this work (Sovon, n.d.). Sovon's State of Birds in the Netherlands 2024 also reports that Wageningen University & Research estimated that volunteers spent at least 641,500 hours in the field recording the occurrence of plants and animals in the Netherlands in 2021. These observations are a major contribution to our knowledge of nature (Sovon, 2024).

Where does the practical difficulty appear?

Counting wildlife properly takes more than pointing a camera at it. Sovon explains that a bird counter needs to recognise the species being counted and follow the right method. Formal surveys use standardised instructions because results collected in different ways are difficult to compare (Sovon, 2024). That matters: a casual photo log can help someone learn, but it should not be treated as equivalent to an official population survey.

For someone who is just getting started, the first hurdle can be more modest: “What am I looking at, how many are there, and how can I keep a record?” While collecting the first photographs for this project, this became very concrete. One image may show a bird that is only a small shape in the water; another may show several birds against grass, reflections or a busy background. Those are ordinary situations for a person taking photos outside, but they are not always easy for an image model.

The problem BirdWatch NL will explore

BirdWatch NL focuses on that beginner step between seeing birds and organising a personal observation. The proposed app lets a user upload a photo, see the model’s predicted bird boxes and category counts, correct mistakes, and save the confirmed observation with a date, general location and notes. The goal is to make observing and learning feel more approachable—not to suggest that a simple app can solve biodiversity loss on its own.

This user problem is a well-grounded design hypothesis, not a claim that interviews have already proved it. The next step is to test it with beginner birdwatchers and find out whether they actually find the process easier.

Problem statement

Beginner birdwatchers may find it difficult to identify and count several common bird groups in one photograph and keep their sightings organised. BirdWatch NL tests whether a custom computer-vision model, combined with a simple observation log and human correction, can make this task easier and more engaging.

Research question

Can a custom Roboflow model help beginner birdwatchers detect and count selected common bird groups in photographs taken in Dutch parks and waterways, and how reliably does it work on photos it has never seen before?

Scope note: The planned labels—ducks, geese, swans, crows, pigeons and seagulls—are broad bird groups, not exact species. The app should make that distinction clear.

2. Sustainable Development Goal (SDG)

Primary SDG: SDG 15 — Life on Land. The United Nations describes this goal as protecting and restoring ecosystems, using them sustainably and halting biodiversity loss. Two targets are especially relevant here: Target 15.1 includes the conservation and sustainable use of inland freshwater ecosystems, while Target 15.5 calls for action to reduce habitat degradation and halt biodiversity loss (United Nations, n.d.).

How BirdWatch NL connects to SDG 15

The connection is indirect, but still meaningful. A person is more likely to pay attention to local nature when they can recognise what they are seeing and keep track of it. By helping beginners look more closely at familiar birds, BirdWatch NL aims to support nature awareness and make personal observations easier to organise. The project also demonstrates how computer vision can be applied to a real-world environmental learning task, rather than only to an abstract technical example.

- Awareness: the tool may help a user become more interested in birds they usually pass without noticing.
- Observation: a saved photo, date, approximate location and note can help someone revisit what they saw and learn over time.
- Innovation: the project trains a custom model using images collected and annotated for this specific task.

What the SDG connection does—and does not—mean

BirdWatch NL will not restore habitats, stop population declines or measure an official SDG indicator. It also will not produce scientific monitoring data simply because it counts birds in a photograph. Formal ecological monitoring needs agreed methods, consistent effort and quality checks. The realistic contribution is smaller: helping people learn about everyday biodiversity and testing whether a beginner-friendly AI workflow is useful. That is a credible educational link to SDG 15, as long as the project does not claim a conservation impact that has not been measured.

3. User group

Primary users: beginner birdwatchers

The main user is someone who enjoys walking through a park or beside a canal but does not yet know many birds confidently. They may take photos with a phone, search for the bird later, or simply leave the photo in their camera roll. This is a proposed user group; it still needs to be checked with real users.

What might this user need?

- A quick first suggestion about which supported bird group appears in the photo.
- Separate boxes around each detected bird and a count for each supported category.
- A simple way to correct wrong boxes, missed birds or incorrect category labels.
- A place to save the date, a general location and a short note about the observation.
- A clear reminder that the model can make mistakes and may not recognise birds outside its training categories.

A realistic use scenario

Imagine a student taking a picture of several birds beside a canal. Instead of needing to count each one from scratch, they upload the image to BirdWatch NL. The model suggests boxes and category labels. The student checks the result, fixes any mistakes and saves the observation. If the app makes the process clearer or faster, that is useful evidence. If the student still has to correct many errors, that is equally important feedback for improving the design.

Secondary users

Teachers, families, students and local nature-education groups could use the prototype for a simple outdoor learning activity. Citizen-science volunteers may also find the idea interesting, but BirdWatch NL should not be marketed as an alternative to expert survey methods or existing scientific databases.

How the user need should be tested

A small usability test with three to five beginners would be a practical start. Ask each person to identify and count birds in a few photos without the app, then repeat the task with BirdWatch NL. Compare the time taken, the number of corrections, count accuracy and how confident the person feels about the result. This would provide more useful evidence than assuming the app helps simply because the AI produces a prediction.

4. Existing solutions and approaches

A. Sovon's bird monitoring programmes

Sovon coordinates structured surveys of breeding birds, urban birds and waterbirds. The results are used to follow population trends and support decisions about nature management. For waterbirds, monthly counts generally take place from September to April. These programmes are important because observations collected with a consistent method can be compared over time (Sovon, n.d.; Sovon, 2024).

How this differs from BirdWatch NL: Sovon's surveys are designed to produce dependable monitoring data. BirdWatch NL is intended for informal learning and personal photo records. It should not replace Sovon's methods, and a saved BirdWatch NL observation should not be presented as an official count.

B. ObsIdentify

ObsIdentify is already an established AI-based nature-identification app. Users can photograph a wild animal, plant or mushroom in Europe or the Dutch Caribbean, receive a species suggestion and collect observations. The app's recognition system draws on millions of observations from users of nature-recording platforms (Observation International, 2026).

What this means for the project: BirdWatch NL cannot claim that it is the first app to identify nature from a photograph. The question is whether a custom, limited-category detector paired with a very simple count-correction and CSV-log workflow helps the particular users this project targets.

C. Merlin Bird ID and eBird Photo ID

Merlin Bird ID helps users identify birds by description, sound and photographs. In eBird's photo workflow, Merlin's model can look for birds in an image, draw a box around each one and assign a possible species label. Cornell Lab's guidance also warns that the model can make mistakes, especially when the bird is hidden or several birds are present; it recommends treating AI suggestions as help rather than a final answer (eBird Support, 2025).

This is a particularly relevant comparison because it already covers much of the technical idea behind BirdWatch NL. The difference is not that BirdWatch NL has invented bird detection. It is a student-built prototype with a deliberately small set of broad categories, an editable count and a straightforward personal log. Whether this is actually easier for beginners remains something to test.

D. What research on wildlife AI tells us

Computer vision can help organise and review animal images, but results depend on the images used to train and test a model. Research comparing image-classification approaches on camera-trap photographs highlights that image framing and cropping can affect performance (Gadot et al., 2024). For BirdWatch NL, the practical lesson is simple: a model that works on close, clear birds should not be assumed to work just as well when birds are small, partly hidden or photographed against a different background.

E. The opportunity for BirdWatch NL

The research points to a narrow but achievable opportunity: create a simple learning tool where a beginner can see which selected bird groups the model detects, check the count, correct errors and save a record. Its value should be judged on actual results: does it detect the selected categories on unseen photos, and does it make the observation task easier? If testing does not show an advantage, the project should report that honestly rather than inventing a unique benefit.

5. Ethical considerations

A. Incorrect predictions can mislead people

The model may miss a bird, mistake a reflection for an animal, draw an inaccurate box or assign the wrong category. A confident-looking result can still be wrong. This matters most if people copy predictions into a record without checking them.

How we can reduce the risk: let users edit or remove predictions, display a clear warning, test the model on photos it has not seen and show examples of mistakes. Do not present the output as a scientific conclusion.

B. Fairness across classes and settings

If the training set contains many ducks but very few crows or swans, the model may perform well overall while being unreliable for the smaller classes. If almost all photos come from one canal, the model may learn the background as much as the bird. A photo of a coot illustrates why class definitions matter: the dark bird with a white bill and forehead shield in the pilot photographs appears to be a Eurasian coot, which belongs to the rail family, not the duck group (Vogelbescherming Nederland, n.d.).

How we can reduce the risk: define the labels before annotating; count the number of images and individual bird boxes per category; collect different angles, distances, locations and light conditions; and decide deliberately how to handle birds outside the target classes. Do not quietly label a coot as a duck just to make the data fit the plan.

C. Privacy and sensitive locations

A photo taken in a public park may accidentally include a child, passer-by, house number or other identifying detail. Some wildlife locations may also be sensitive, particularly nests or rare species. For common birds in a public canal, the risk is usually limited, but it should still be considered.

How we can reduce the risk: avoid photographing identifiable people, crop or blur them when necessary, use a general location rather than exact coordinates when precision is not needed, and never share precise nest locations publicly.

D. Cloud processing and API keys

If the app sends an uploaded image to a hosted Roboflow model for analysis, users should know that the photo leaves their device for processing. The app should explain what information is saved and where. API keys must be kept out of public GitHub repositories and source code; use a local environment variable or an appropriate secrets setting instead.

E. Respect for wildlife and photo ownership

The camera should not become a reason to disturb birds. Take photos from a respectful distance, stay on paths where possible, and do not approach nests or deliberately frighten birds to get a better image. For the training set, use the self-collected photographs required by Hackathon 7. Do not count images downloaded from the internet as your own photographs, and get permission before using somebody else's image.

F. Keep a person responsible for the final record

BirdWatch NL should suggest, not decide. Users should be able to reject a prediction and save only the record they have reviewed. The ethical goal is not to make the AI look perfect; it is to make its limits visible and keep the human in control.

6. Feasibility and direct link to Hackathon 7

Why BirdWatch NL fits Eyes on Earth

Hackathon 7 focuses on computer vision: collecting images, training a custom model with Roboflow, and demonstrating that it works on real examples. BirdWatch NL fits that brief because the AI model is central to the idea. The app is not just a gallery of bird photographs; it needs to locate individual birds in an image, assign each detected bird to a defined category and turn those predictions into a count that a user can check.

This also gives the SDG connection a concrete form. The project uses the required computer-vision tool for a real nature-observation task, links to SDG 15 through environmental learning and awareness, and includes an ethical reflection about prediction errors, data quality, privacy and the limits of the claims. The model must still be trained and evaluated properly; simply connecting a pre-existing AI service would not demonstrate the same work.

A realistic build plan

1. Collect at least 300 photographs taken by the project team, following the Hackathon 7 brief. Use different days, distances, angles, group sizes, backgrounds and lighting conditions. Do not substitute internet images for the required self-collected photos.
2. Start with a small pilot of around 30 photos. Check which bird categories appear often enough and whether each label can be applied consistently. If six classes are too difficult to balance, narrow the scope rather than pretending that every category will be equally reliable.
3. Define each class in writing before annotation. For object detection, draw a box around each visible bird in a supported class and apply the same label consistently. Track both image totals and the number of labelled birds per category.
4. Create training, validation and test splits. Keep near-duplicate photos and images from the same short encounter in the same split where possible, so the final test is not inflated by almost identical photographs appearing on both sides.
5. Train the model in Roboflow and review the results per class. Record examples of missed birds, incorrect labels and false detections, not just one overall score.
6. Connect the trained model to the Streamlit app. Show the predictions and counts, allow people to correct the result, and save the reviewed observation with its date, general location and notes.
7. Test the complete app with photographs that were not used in training. Include difficult cases such as distant birds, several birds together, reflections, partial obstruction and backgrounds from a different place.

What would count as success?

A successful upload or one correct prediction proves that part of the application works; it does not prove the model is accurate. A more convincing evaluation would report:

- Detection results for each category, including precision and recall or the metrics supplied by Roboflow.
- Counting error: compare the true number of birds in each test photo with the model's predicted count.
- A short error review with examples of what the model misses and where it struggles.
- A small beginner usability test to see whether people can review, correct and save an observation without confusion.

Important scope decision: species or bird groups?

The six planned labels are broad groups rather than exact species. The app should say "duck", for example, only if that is the actual training label; it should not claim to recognise a mallard specifically unless mallard is a separate class. The initial canal photos also include birds that appear to be coots. Decide whether to add coot as a class, define an "other bird" category if the training plan supports it, or leave unsupported birds out of the labelled dataset and clearly note that they are outside the current scope.

Feasibility assessment

The prototype is technically realistic for a student project because the workflow can be kept small: a focused set of categories, one Roboflow detector, a simple Streamlit interface and a CSV observation log. The hardest part is not writing the upload button; it is collecting a varied dataset, annotating it consistently and being honest about the mistakes. The project description reports one successful end-to-end test with a seagull image, which is a useful first milestone. It is not enough to establish overall accuracy, so further testing is essential.

A realistic success criterion is: "BirdWatch NL can detect and count selected bird groups on a defined set of unseen photographs, shows users where its predictions may be wrong, and lets them correct and save the result." That is specific enough for Hackathon 7 and testable within a student prototype.

7. Conclusion

BirdWatch NL has a meaningful starting point because bird observations matter, and collecting them carefully takes both knowledge and effort. Dutch monitoring programmes already rely on volunteers, and their data help researchers and policymakers understand how bird populations change. At the same time, official trends show that the picture is mixed: some bird groups are doing better, while others—especially many farmland birds—have declined. This is why the project should talk about the importance of observation, rather than claiming that all birds are disappearing.

The research also shows that BirdWatch NL needs to be honest about the solutions that already exist. Apps such as ObsIdentify and Merlin already help people identify nature from images, and Merlin can detect multiple birds in photos. BirdWatch NL's opportunity is therefore not to claim it is the first or best bird-identification app. It is to build and test a small, custom model for selected bird groups, combine it with a clear correction-and-counting workflow, and see whether beginners find that useful.

This is a strong fit for AI for Good — Hackathon 7: Eyes on Earth because the project applies a custom computer-vision model to a real environmental learning task and connects it to SDG 15. Its contribution is modest but defensible: it may help people notice local birds, learn from their photographs and organise personal observations. Whether it actually saves time or improves counting must be demonstrated through testing. It should never be presented as a replacement for professional monitoring or as proof of conservation impact.

References

The sources below support the problem context, existing tools, SDG connection and proposed testing approach.

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